Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Welcome to the Course

- Welcome to Applied Unsupervised Learning in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 5 min
- Course Syllabus
  Reading • 10 min
- Additional Resources
  Reading • 10 min
- Meet Your Fellow Learners
  Discussion Prompt • 15 min
- Help Us Learn About You
  Reading • 5 min

Lesson 1: Methods for Dimensionality Reduction

- Dimensionality Reduction: A Brief Introduction
  Video • 16 min
- Dimensionality Reduction with Feature Selection: Information Gain
  Video • 15 min
- Time to Practice: Dimensionality Reduction
  Graded Assignment • Grade: 100%
- Dimensionality Reduction with Feature Selection: Principal Component Analysis (PCA) Explained
  Video • 20 min
- Visualizing PCA Results: Foundations
  Video • 8 min
- Visualizing PCA Results: Biplots and Variance Plots
  Video • 17 min
- Ten Quick Tips for Effective Dimensionality Reduction
  Reading • 45 min
- Time to Practice: Principal Component Analysis (PCA)
  Graded Assignment • Grade: 100%
- Singular Value Decomposition (SVD)
  Video • 21 min
- Applications of SVD in Data Science
  Video • 13 min
- Time to Practice: Singular Value Decomposition (SVD)
  Graded Assignment • Grade: 100%
- Manifold Learning: Multidimensional Scaling (Part 1)
  Video • 17 min
- Manifold Learning: Multidimensional Scaling (Part 2)
  Video • 11 min
- Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2)
  Graded Assignment • Grade: 100%
- Manifold Learning: t-Distributed Stochastic Neighbor Embedding (t-SNE)
  Video • 13 min
- Manifold Learning: Uniform Manifold Approximation and Projection (UMAP)
  Video • 6 min
- Time to Practice: Manifold Learning (t-SNE, and UMAP)
  Graded Assignment • Grade: 100%

Lesson 2: Density Estimation

- Density Estimation Part 1: Probability Density Functions
  Video • 12 min
- Density Estimation Part 1: Parametric vs. Non-Parametric Density Estimator
  Video • 13 min
- Time to Practice: Density Estimation Methods (Part 1)
  Graded Assignment • Grade: 100%
- Density Estimation Part 2: Local Density Estimators
  Video • 11 min
- Density Estimation Part 2: Kernel Density Estimators
  Video • 15 min
- Density Estimation Part 2: Evaluating Density Estimators
  Video • 4 min

Introduction to Module 1 Programming Assignment: An Introduction to Unsupervised Learning

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For the Module 1 programming assignment, you will focus on dimensionality reduction with PCA, using a music dataset of Spotify tracks.

Datasets often have a massive number of different features for each data point. In the case of the Spotify tracks dataset you will use, you will have features for each song, such as popularity, duration, and "loudness." Applications that utilize data like this—in this case, recommendation systems in music streaming apps—often need to reduce the number of features/dimensions to the data to make comparing each data point, or song in this case, computationally efficient.

In this assignment, you will use PCA dimensionality reduction on a dataset of music tracks and look into some valuable tasks you can perform with the resulting PCA object.

1. **First**, you will create a PCA object on the song dataset and examine key attributes of the resulting object.
2. **Second**, with the PCA object you created in the first step, you will create a scree plot, which can help determine the number of principal components needed to account for an adequate amount of variance in the data.
3. **Third**, you will use a biplot to visualize both the dataset and its features (variables), which will allow you to get insight into relationships between song features.
4. **Fourth**, you will estimate a probability density over songs using kernel density estimation. This will allow you to glean interesting facts about songs and their relationships. In this step, you will compute the likelihood of songs in the dataset to see how 'typical' or 'atypical' they are in the dataset.

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